Abbreviations

Ø	DIAMETER	HGT	HEIGHT
#/FT	PER FOOT	HORIZ	HORIZONTAL
@	AT	HSA	HEADED STUD ANCHOR
		HSS	HOLLOW STEEL SECTION
AB	ANCHOR BOLT		
ABV	ABOVE	IAW	IN ACCORDANCE WITH
ADJ	ADJACENT	IBC	INTERNATIONAL BUILDING CODE
AFF	ABOVE FINISHED FLOOR	IF	INSIDE FACE
AISC	AMERICAN INSTITUTE OF STEEL CONSTRUCTION	INT	INTERIOR
ALT	ALTERNATE		
APPROX	APPROXIMATE	JT	JOINT
ARCH	ARCHITECT		
ASCE	AMERICAN SOCIETY OF CIVIL ENGINEERS	KSI	KIPS PER SQUARE INCH
ASTM	AMERICAN SOCIETY FOR TESTING AND MATERIALS		
		LLH	LONG LEG HORIZONTAL
B/	BOTTOM OF	LLV	LONG LEG VERTICAL
BC	BOTTOM CHORD	LSV	LONG SIDE VERTICAL
BLDG	BUILDING		
BLW	BELOW	MAX	MAXIMUM
BM	BEAM	MC	MISCELLANEOUS CHANNEL
BRG	BEARING	MECH	MECHANICAL
BTWN	BETWEEN	MFR	MANUFACTURER
		MIN	MINIMUM
C	CHANNEL	MISC	MISCELLANEOUS
CFMF	COLD-FORMED METAL FRAMING	MO	MASONRY OPENING
CIP	CAST-IN-PLACE	MOD	MODIFIED
CJ	CONTROL JOINT	MTL	METAL
CL	CENTER LINE		
CLR	CLEAR	NIC	NOT IN CONTRACT
CMU	CONCRETE MASONRY UNIT	NTS	NOT TO SCALE
COL	COLUMN		
CONC	CONCRETE	OAE	OR APPROVED EQUAL
CONT	CONTINUOUS	OC	ON CENTER
		OPNG	OPENING
DBA	DEFORMED BAR ANCHOR		
DBL	DOUBLE	PAF	POWDER ACTUATED FASTENER
DET	DETAIL	PL	PLATE
DIA	DIAMETER	PSF	POUNDS PER SQUARE FOOT
DIAG	DIAGONAL	PSI	POUNDS PER SQUARE INCH
DWG	DRAWING		
		RAD	RADIUS
E/	EDGE OF	REINF	REINFORCED
EA	EACH	REQD	REQUIRED
EB	EXPANSION BOLT		
EFOB	EXTERIOR FACE OF BRICK	SIM	SIMILAR
EIFS	EXTERIOR INSULATION FINISHING SYSTEM	SLV	SHORT LEG VERTICAL
EJ	EXPANSION JOINT	SQ	SQUARE
EL	ELEVATION	SS	STAINLESS STEEL
EOD	EDGE OF DECK	STD	STANDARD
EOS	EDGE OF SLAB	STL	STEEL
EQ	EQUAL	STRUCT	STRUCTURAL
ES	EACH SIDE		
EW	EACH WAY	T/	TOP OF
EXP	EXPANSION	THK	THICKNESS
EXT	EXTERIOR	TRANS	TRANSVERSE
		t/s	TOP OF STEEL
F/	FACE OF	TYP	TYPICAL
FAB	FABRICATE		
FDN	FOUNDATION	UNO	UNLESS NOTED OTHERWISE
FFE	FINISH FLOOR ELEVATION		
FIN	FINISH	VERT	VERTICAL
FLR	FLOOR	VIF	VERIFY IN FIELD
FOS	FACE OF STUD		
FTG	FOOTING	W/	WITH
FV	FIELD VERIFY	WF	WIDE FLANGE
		WP	WORK POINT
GA	GAUGE	WWF	WELDED WIRE FABRIC
GALV	GALVANIZED		
GC	GENERAL CONTRACTOR		
GR	GRADE		

EXTERIOR SCHEDULE OF METAL STUD SIZES
115 MPH, EXPOSURE C, Fy = 33 ksi

WALL HEIGHT	6" STUD		8" STUD	
WIND ZONE 4	L/360	L/600	L/360	L/600
Up to 11'-0"	600S 162-43 @ 16" o/c	600S 162-43 @ 16" o/c	800S 162-43 @ 16" o/c	800S 162-43 @ 16" o/c
Over 11'-0" to 12'-6"	600S 162-43 @ 16" o/c	600S 162-68 @ 16" o/c	800S 162-54 @ 16" o/c	800S 162-54 @ 16" o/c
Over 12'-6" to 15'-6"	600S 162-68 @ 16" o/c	600S 200-97 @ 16" o/c	800S 162-54 @ 16" o/c	800S 162-54 @ 16" o/c
Over 15'-6" to 16'-6"	600S 162-97 @ 16" o/c	600S 200-97 @ 12" o/c	800S 162-54 @ 16" o/c	800S 162-68 @ 16" o/c
Over 16'-6" to 20'-0"	600S 200-97 @ 12" o/c	N/A	800S 162-68 @ 16" o/c	800S 250-97 @ 16" o/c
Over 20'-0" to 21'-0"	600S 250-97 @ 12" o/c	N/A	800S 162-97 @ 16" o/c	800S 200-97 @ 12" o/c
WIND ZONE 5	L/360	L/600	L/360	L/600
Up to 11'-0"	600S 162-43 @ 16" o/c	600S 162-54 @ 16" o/c	800S 162-54 @ 16" o/c	800S 162-54 @ 16" o/c
Over 11'-0" to 12'-6"	600S 162-54 @ 16" o/c	600S 162-68 @ 16" o/c	800S 162-54 @ 16" o/c	800S 162-54 @ 16" o/c
Over 12'-6" to 15'-6"	600S 162-97 @ 16" o/c	600S 250-97 @ 16" o/c	800S 162-54 @ 16" o/c	800S 162-68 @ 16" o/c
Over 15'-6" to 16'-6"	600S 162-97 @ 16" o/c	600S 200-97 @ 12" o/c	800S 162-54 @ 16" o/c	800S 162-97 @ 16" o/c
Over 16'-6" to 20'-0"	600S 200-97 @ 12" o/c	N/A	800S 162-97 @ 16" o/c	800S 250-97 @ 16" o/c
Over 20'-0" to 21'-0"	N/A	N/A	800S 162-97 @ 16" o/c	800S 200-97 @ 12" o/c

NOTES:
1. WHERE EXTERIOR FINISH IS EXTERIOR FINISHES, EFIS OR METAL PANEL USE DEFLECTION LIMITATION OF L/360.
2. WHERE EXTERIOR FINISH IS MASONRY OR BRICK USE DEFLECTION LIMITATION OF L/600.

STRUCTURAL SUBMITTALS

SPECIFICATION SECTION	SUBMITTAL	SUBMITTAL TYPE		DESIGN BY CONTRACTOR'S LICENSED ENGINEER
		ACTION	INFORMATIONAL	
03 1000 - CONCRETE FORMING AND ACCESSORIES	PRODUCT DATA	•		
	FORMWORK SHOP DRAWINGS (ELEVATED CONCRETE ONLY)	•		•
03 2000 - CONCRETE REINFORCING	PRODUCT DATA	•		
	REINFORCING SHOP DRAWINGS	•		
	CONSTRUCTION JOINT LAYOUT	•		
	REINFORCING MATERIAL CERTIFICATES		•	
	REINFORCING MATERIAL TEST REPORTS		•	
03 3000 - CAST-IN-PLACE CONCRETE	PRODUCT DATA	•		
	CONCRETE DESIGN MIXTURES	•		
	SLAB-ON-GRADE JOINT LAYOUT	•		
	MATERIAL CERTIFICATES		•	
	MATERIAL TEST REPORTS		•	
04 2200 - CONCRETE UNIT MASONRY	PRODUCT DATA	•		
	SHOP DRAWINGS	•		
	MATERIAL CERTIFICATES		•	
	MIX DESIGNS FOR MORTAR AND GROUT		•	
	STATEMENT OF COMPRESSIVE STRENGTH FOR MASONRY		•	
05 1200 - STRUCTURAL STEEL FRAMING	PRODUCT DATA	•		
	SHOP DRAWINGS	•		
	CONNECTION DESIGN CALCULATIONS	•		•
	PAINT COMPATIBILITY CERTIFICATES		•	
	MILL TEST REPORTS		•	
	PRODUCT TEST REPORTS		•	
05 2100 - STEEL JOIST FRAMING	STEEL JOIST SHOP DRAWINGS	•		
05 3100 - STEEL DECKING	STEEL DECKING PRODUCT DATA & SHOP DRAWINGS	•		
05 4000 - COLD-FORMED METAL FRAMING	PRODUCT DATA	•		
	SHOP DRAWINGS	•		
	EXTERIOR METAL STUD DESIGN CALCULATIONS	•		•
	LOAD-BEARING METAL STUD DESIGN CALCULATIONS	•		
	PRODUCT CERTIFICATES		•	
	PRODUCT TEST REPORTS		•	
05 5113 - , HAND RAILS, & GUARDS	PRODUCT DATA	•		
	SHOP DRAWINGS	•		
	DESIGN CALCULATIONS	•		•
	PAINT COMPATIBILITY CERTIFICATES		•	

MINIMUM REINFORCED CONCRETE STRENGTHS
F_c AT 28 DAYS (U.N.O. ON SCHEDULES)
(TYPE 1 CEMENT ONLY. DO NOT SUBSTITUTE FLYASH OR SLAG FOR CEMENT)

LOCATION	DENSITY	STRENGTH
SLAB-ON-GRADE	145 PCF	3000 PSI
FOUNDATIONS AND PIERS	145 PCF	3000 PSI
SLAB ON GRADE AT BIN-SUPPORTED MEZZANINE	145 PCF	5000 PSI
FRAMED SLABS IN COMPOSITE STEEL CONSTRUCTION	115 PCF	4000 PSI
FRAMED SLABS IN BAR JOIST FLOOR CONSTRUCTION	145 PCF	3500 PSI
STAIR TREADS AND LANDINGS	145 PCF	3000 PSI
BASEMENT, TIEBACK, AND RETAINING WALLS	145 PCF	3000 PSI
NON-SHRINK GROUT AT STEEL COLUMN BASEPLATES	145 PCF	5000 PSI

MINIMUM LAP SPLICE LENGTH SCHEDULE

BAR TYPE	3000 PSI CONC.								
	#3	#4	#5	#6	#7	#8	#9	#10	#11
TOP BARS	29"	38"	47"	56"	82"	94"	105"	118"	131"
OTHER BARS	22"	29"	36"	43"	62"	72"	81"	91"	101"
BAR TYPE	4000 PSI CONC.								
	#3	#4	#5	#6	#7	#8	#9	#10	#11
TOP BARS	25"	33"	40"	48"	70"	81"	91"	103"	113"
OTHER BARS	20"	25"	31"	38"	55"	62"	70"	79"	87"

MINIMUM PRETENSION OF BOLTS (KIPS)

BOLT DIAMETER	A325 BOLTS	A490 BOLTS
3/4" DIAMETER	28	35
7/8" DIAMETER	39	49
1" DIAMETER	51	64

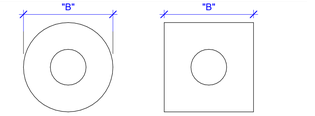
NON-LOAD-BEARING BRICK LINTEL SCHEDULE

SPAN	LINTEL SIZE
≤ 2'-0"	1 FLAT PLATE - 3-1/2" x 3/8"
2'-0" TO 5'-0"	1 ANGLE - 3-1/2" x 3-1/2" x 5/16"
5'-0" TO 6'-0"	1 ANGLE - 4" x 3-1/2" x 5/16" S.L.O.
6'-0" TO 7'-0"	1 ANGLE - 5" x 3-1/2" x 5/16" S.L.O.
7'-0" TO 8'-0"	1 ANGLE - 5" x 3-1/2" x 3/8" S.L.O.
8'-0" TO 9'-0"	1 ANGLE - 6" x 4" x 3/8" S.L.O.
9'-0" TO 10'-0"	1 ANGLE - 8" x 4" x 7/16" S.L.O.

NOTES:
1. SEE SCHEDULE FOR STEEL GRADE AND STRENGTH
2. ANGLES ARE FOR 4" BRICKWORK
3. BEAR ALL LINTELS A MINIMUM OF 8" AT EACH END
4. ANGLES SHALL BE PLACED SHORT LEG OUT (S.L.O.)
5. FOR PAINTING OR GALVANIZING OF ANGLES, SEE ARCHITECT

MINIMUM ANCHOR BOLT WASHER DIMENSIONS

ANCHOR BOLT DIAMETER	MINIMUM WASHER "B" DIMENSION	MINIMUM WASHER THICKNESS
3/4"	2"	1/4"
7/8"	2-1/2"	5/16"
1"	3"	3/8"
1-1/8" OR 1-1/4"	3"	1/2"
1-3/8" OR 1-1/2"	3-1/2"	1/2"
1-5/8" OR 1-3/4"	4"	5/8"
1-7/8" OR 2"	5"	3/4"
2-1/4" OR 2-1/2"	5-1/2"	7/8"



CIRCULAR WASHER SQUARE WASHER

STRUCTURAL STEEL STRENGTHS

STEEL SHAPE	ASTM	F_y (ksi)	F_u (ksi)
ANGLES, PLATES, S SHAPE, AND MISC., CHANNELS MC15 AND SMALLER, AND CHANNELS C8 AND SMALLER	A36	36	58
W SHAPES, MISC CHANNELS MC18 AND LARGER, CHANNELS C10 AND LARGER	A992	50	65
HSS RECTANGULAR AND SQUARE	A500 GRADE C	50	62
HSS ROUND	A500 GRADE C	46	62
PIPE	A53 GRADE B	35	60
SHEAR STUDS	A108	-	65
ANCHOR RODS	F1554 GRADE 36	36	58
HIGH STRENGTH BOLTS 3/4" TO 1" Ø 1-1/8" TO 1-1/2" Ø 1-1/8" TO 1-1/2" Ø AT SE CONN.	F3125 A325 F3125 A325 F3125 A490	- - -	120 105 150
WELDED WIRE FABRIC 6x6-W.9xW2.9	A185	-	-
CONCRETE REINFORCING	A615 GRADE 60	60	-
CONCRETE REINFORCING STIRRUPS, TIES, AND HOOPS	A615 GRADE 40	40	-

DESIGN NOTES

REFERENCE CODES

INTERNATIONAL BUILDING CODE	IBC 2024 w/ Georgia Amendments
MINIMUM DESIGN LOADS	ASCE 7-22
STRUCTURAL CONCRETE	ACI 318-19
STRUCTURAL STEEL	AISC 360-22
STRUCTURAL STEEL SEISMIC	AISC 341-22
STRUCTURAL MASONRY	TMS 402-16
COLD-FORMED STEEL	AISI S100-16

LIVE LOADS

OCCUPANCY OR USE	UNIFORM (PSF)	CONC. (LBS)	OCCUPANCY OR USE	UNIFORM (PSF)	CONC. (LBS)
ROOF	20	300	HANDRAILS AND GUARDS GRAB BARS		200 250

ROOF SNOW LOAD

FLAT-ROOF SNOW LOAD (p_f)	6 PSF	SNOW IMPORTANCE FACTOR (I_s)	1.0
SNOW EXPOSURE FACTOR (C_e)	1.0	THERMAL FACTOR (C_t)	1.15
DESIGN FROST DEPTH	12"		

ROOF RAIN LOAD

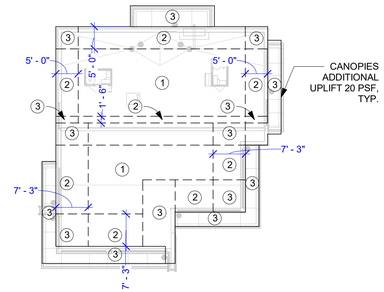
RAIN INTENSITY (i)	7.39 INCHES/HOUR
DESIGN EVENT	15-MINUTE/100-YEAR RETURN PERIOD

WIND DESIGN CRITERIA

DESIGN WIND SPEED ULTIMATE (V) NOMINAL (V_{asd})	 107 MPH 83 MPH	INT. PRESSURE COEFFICIENT	±0.18
RISK CATEGORY	2		
WIND EXPOSURE (X-X)	C	COMPONENTS AND CLADDING WIND PRESSURES: SEE TABLES, THIS SHEET	
WIND EXPOSURE (Y-Y)	C		

SEISMIC DESIGN CRITERIA

RISK CATEGORY	2
SEISMIC IMPORTANCE FACTOR	I_e = 1.0
MAPPED SPECTRAL RESPONSE ACCELERATIONS	S_s = 0.230 g S_1 = 0.084 g
SITE CLASS	D
MAPPED SPECTRAL RESPONSE ACCELERATIONS	S_{DS} = 0.190 g S_{D1} = 0.120 g
SEISMIC DESIGN CATEGORY	B
BASIC SEISMIC FORCE-RESISTING SYSTEM(S)	STEEL SYSTEMS NOT SPECIFICALLY DETAILED FOR SEISMIC RESISTANCE
DESIGN BASE SHEAR	$F_x = F_y$ = 4 k
RESPONSE MODIFICATION FACTOR(S)	R = 3
ANALYSIS PROCEDURE	EQUIVALENT LATERAL FORCE

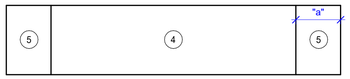


COMPONENTS AND CLADDING ROOF (ENCLOSED) 107 MPH EXP. "C" 'ULT' EXTERNAL PRESSURES (PSF)				
A_e (EFFECTIVE AREA)	ZONE 1'	ZONE 1	ZONE 2	ZONE 3
A_e = 10 SQ. FT.	+16.0, -23.2	+16.0, -40.3	+16.0, -53.2	+16.0, -72.5
A_e = 20 SQ. FT.	+16.0, -30.7	+16.0, -38.2	+16.0, -49.9	+16.0, -66.0
A_e = 50 SQ. FT.	+16.0, -26.4	+16.0, -33.9	+16.0, -44.6	+16.0, -57.0
A_e = 100 SQ. FT.	+16.0, -23.2	+16.0, -31.7	+16.0, -42.4	+16.0, -50.4

COMPONENTS AND CLADDING WALL (ENCLOSED) 107 MPH EXP. "C" 'ULT' EXTERNAL PRESSURES (PSF)			
A_e (EFFECTIVE AREA)	ZONE 4	ZONE 5	
A_e = 10 SQ. FT.	+23.2, -25.1	+23.2, -30.9	
A_e = 20 SQ. FT.	+22.1, -24.1	+22.1, -28.8	
A_e = 50 SQ. FT.	+20.8, -22.7	+20.8, -26.1	
A_e = 100 SQ. FT.	+19.7, -21.7	+19.7, -24.1	

NOTES:
1. *INTERPOLATION MAY BE UTILIZED FOR EFFECTIVE AREAS THAT OCCUR BETWEEN VALUES SHOWN ON THE TABLE.*
2. *PLUS AND MINUS SIGN INDICATES THE PRESSURE ACTING TOWARD AND AWAY FROM THE SURFACES, RESPECTIVELY.*
3. *FORCES AND DIAGRAMS ARE BASED ON IBC / ASCE 7.*

End zone width, a = 5' - 2"



WALL ELEVATION

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GEORGIA REGISTERED No. SE001277 STRUCTURAL ENGINEER MARK W. NANNIS

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